Enhancing Communication in Speech Therapy: Exploring the Cognitive Synergy Between Gesture and Speech

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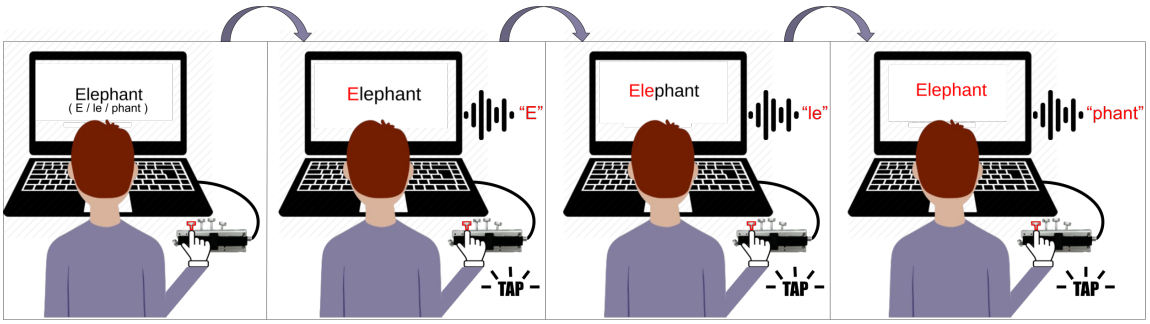

Fig. 1. The user interacts with the interface by tapping a button on the Dextrain Manipulandum, aligning finger movements with vocal articulation. The word "Elephant" is displayed on the screen and is progressively highlighted in red syllable by syllable each time the button is pressed ("E," "le," "phant").

This paper examines the adaptation of a rhythm-based interface, originally designed for manual dexterity rehabilitation, for use in speech therapy. The interface allows users to control synthesized vocal phrases through finger tapping, leveraging the cognitive link between gesture and speech. Through interviews with four therapists and pilot tests with one speech therapist and eight children with speech impairments (autism, Down syndrome, verbal apraxia, dyslexia), we found that the interface improves motivation and therapeutic outcomes by facilitating more interactions between verbally challenged patients and the therapist. Our findings suggest that rhythmic gestures can enhance verbal communication, offering potential for broader therapeutic and educational applications.

CCS Concepts: • **Human-centered computing** → *Auditory feedback*; *User studies*.

Additional Key Words and Phrases: speech therapy, gesture-speech synergy, rhythm, rehabilitation

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1 INTRODUCTION

Rhythm is vital in a variety of rehabilitation therapies, particularly for enhancing movement coordination and synchronization [52]. In physical therapy, rhythmic exercises are crucial for improving motor skills and body awareness, especially for patients recovering from brain injuries or strokes [16]. Similarly, rhythm plays a key role in speech therapy, where it aids in improving fluency, intonation, and articulation [55].

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Our research investigates the cognitive link between hand gestures, rhythm, and speech through an interactive system that allows users to control the rhythm of a synthesized voice via finger tapping [29]. Originally developed for stroke patients to improve manual dexterity [50, 51], the interface demonstrates how integrating hand gestures with speech rhythm can enhance both performance and motivation.

We aim to adapt this system for broader applications. As a preliminary step, we conducted semi-structured interviews with therapists, including two physical therapists and two speech therapists. We found speech therapy particularly promising, given that hand gestures often accompany verbal communication [2]. We then explored how rhythmic gestures might aid in rehabilitating speech disorders by gathering user feedback on the interface with eight children who have various speech disorders [8, 9].

Our results indicate that the interface effectively links motor skills with speech, improving therapy outcomes and patient motivation. Therapists noted that using gestures in combination with rhythms helped children better memorize and pronounce words. The research also highlighted the need for a customizable interface, leading to refinements that personally adapted for each patient.

This interface, initially designed for motor rehabilitation, was shown to be effective in speech therapy, demonstrating that the connection between gestures and speech is beneficial. The technology shows potential for broader therapeutic applications, suggesting that rhythmic integration in therapy could significantly improve patient outcomes in both motor and speech rehabilitation [10]. This research opens the door for further exploration and the development of adaptable therapeutic tools [35].

2 RELATED WORK

2.1 The Role of Rhythm in Rehabilitation

Rhythm is central in motor rehabilitation, improving timing, coordination, and motor learning in neurological patients of strokes or Parkinson's syndrome [4, 36, 38, 45, 53]. By engaging brain regions such as the premotor cortex and supplementary motor area, rhythmic interventions enhance cortical plasticity and motor synchronization, leading to better motor recovery [6, 22, 43]. Repetitive rhythmic patterns accelerate motor learning by incorporating timing into neural circuits [5, 11, 18, 19]. In speech therapy, rhythm aids fluency and coordination [40], with metronome-based interventions improving speech production, particularly in children with verbal apraxia [31]. Rhythm provides a temporal framework that helps patients better organize speech patterns, improving articulation and fluency [27, 39, 40]. Rhythmic tools like Rhythmic Auditory Stimulation (RAS) and the Dextrain Manipulandum [50] are effective in motor recovery [10, 17]. RAS is a therapeutic technique that uses rhythmic auditory cues, such as metronome beats, to improve movement timing and coordination, often used in gait training for neurological patients. The Dextrain Manipulandum is a hardware device designed to measure the pressure and movement of each finger individually, facilitating exercises that improve manual dexterity, finger strength, and temporal precision through fine motor control tasks [51]. Metronome-based systems, which provide rhythmic cues, are also useful in treating speech disorders like stuttering and verbal apraxia, helping organize speech sequences and improve fluency [31].

2.2 How Gestures and Speech are Connected

Research shows a strong cognitive link between gestures and speech [25, 34, 54], where gestures are an integral part of language production [20, 24, 26]. The "Growth Point Theory" posits that gestures and speech arise from a shared cognitive framework, with gestures representing visuospatial aspects of linguistic content [33]. Neuropsychological

evidence supports this connection, indicating overlapping neural circuits for speech and gestures, treating them as part of the same communicative act [13, 49]. Gesture-based therapies have been successfully used in treating neurogenic communication disorders like aphasia [48].

2.3 New Therapies Combining Rhythm and Gestures

Combining rhythm with gesture-based therapies has significant potential for speech rehabilitation, particularly in patients with speech disorders. Studies show that rhythmic synchronization with gestures enhances word recall and pronunciation in these populations [14, 37]. This approach leverages the brain's natural multimodal integration, where motor actions (gestures) and linguistic elements (speech) are processed together, enhancing learning and retention [14, 41]. Gesture-based interfaces, which enhance motivation and engagement through interactive therapy, have shown promise for patients with autism and Down syndrome [12, 23]. Combined with rhythmic repetition, they also improve articulation and fluency by supporting cognitive and motor functions [14]. These results highlight the importance of integrating both rhythm and gestures in therapeutic interventions to effectively support speech rehabilitation.

While current research shows cognitive and neurological benefits of combining rhythm and gestures in rehabilitation, there are significant opportunities for future development. Gesture-rhythm tools could be expanded in educational and therapeutic contexts, with future research focusing on more interactive, adaptive gesture-speech interfaces across a wider range of conditions.

3 DEVELOPMENT & DESCRIPTION OF THE INTERFACE

3.1 Interface Design

Our system is built as an application for the Dextrain Manipulandum (DM) [50], a device that accurately measures the force exerted by each finger, originally designed for rehabilitating manual dexterity in stroke patients. In this study, the DM is adapted to assist users in improving both their speech and fine motor skills through rhythmic repetition exercises (see figure 5). The platform allows users to replicate words and phrases they hear, with a focus on rhythmic accuracy and articulation. The system combines the DM with a Unity program and the FMOD audio interface [42] to generate synthesized speech segments, ensuring precise timing and optimal sound clarity.

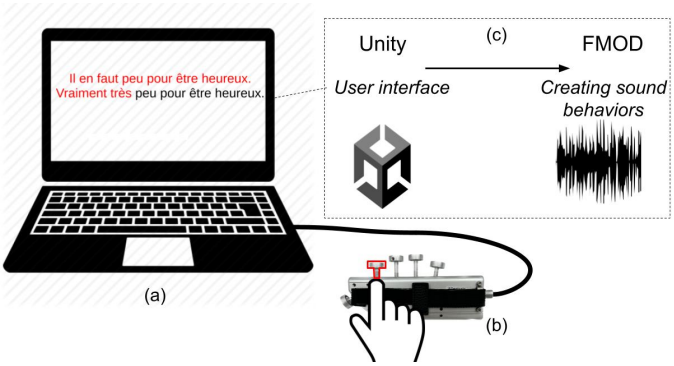


Fig. 2. Illustration of the interface system. (a) The user interface displays black words on the screen, with each syllable turning red as the user presses a button, accompanied by an audible output. (b) The Dextrain Manipulandum [50] is used for tapping, where the participant presses a piston with their index finger, recording precise movements and timing. (c) The software, developed in Unity with FMOD, controls the synthesis of vocal and musical sounds.

3.2 Specific Features

User Interface and Hardware: Users see the words they need to reproduce displayed on the screen in black. With each button pressed by their index finger, a syllable of the word is audibly generated, and the corresponding syllable on the screen turns red. This allows users to visually and audibly track their progress, reinforcing the association between movement, sound, and speech.

Sound Synthesis and Control: The sound is generated using pre-recorded audio files, with each word segmented into syllables with precise timestamps. When a user presses the button, the corresponding syllable is played, helping to maintain a consistent rhythm and practice correct pronunciation. This method ensures that users can focus on synchronizing their movements with sound production, thereby improving their rhythmic accuracy and articulation.

Data Preparation: To prepare the speech content for use, the moments of transition between different phonemes need to be labeled accurately. To achieve precise timing for the speech exercises, we developed a semi-automatic pipeline for processing vocal tracks in .wav format (see figure 3). First, we separate the vocals from the instrumental background using Vocal Remover [1], an online AI-based tool. Next, we segment the phonemes using Montréal Forced Aligner [32], which generates a Praat TextGrid file containing the start and end timestamps of each phoneme [7]. We manually verify and correct these timestamps in Praat to ensure accurate synchronization (see figure 4). Finally, a Python script converts the TextGrid to the required format for our interface. The correctly labeled speech segments are then integrated into the exercises.

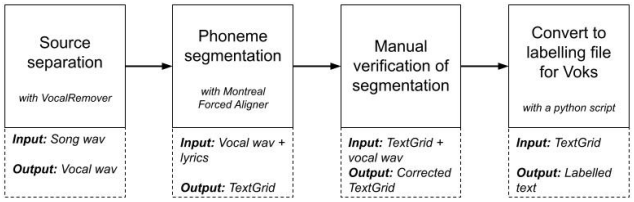
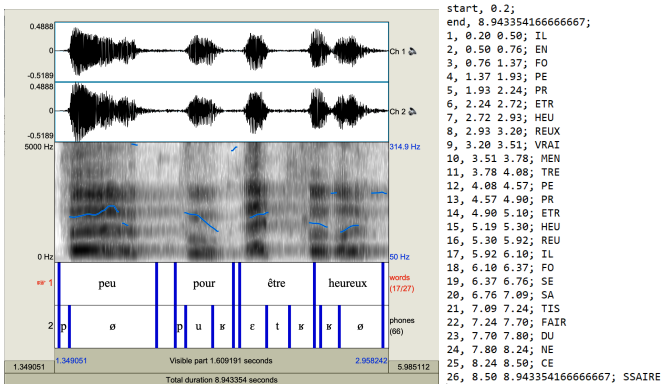


Fig. 3. Data preparation pipeline

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end, 8.943354166666667;
1, 0.20 0.50; IL
2, 0.50 0.76; EN
3, 0.76 1.37; FO
4, 1.37 1.93; PE
5, 1.93 2.24; PR
6, 2.24 2.72; ETR
7, 2.72 2.93; HEU
8, 2.93 3.20; REUX
9, 3.20 3.51; VRAI
10, 3.51 3.78; MEN
11, 3.78 4.08; TRE
12, 4.08 4.57; PE
13, 4.57 4.90; PR
14, 4.90 5.10; ETR
15, 5.19 5.30; HEU
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17, 5.92 6.10; IL
18, 6.10 6.37; FO
19, 6.37 6.76; SE
20, 6.76 7.09; SA
21, 7.09 7.24; TIS
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24, 7.80 8.24; NE
25, 8.24 8.50; CE
26, 8.50 8.943354166666667; SSAIRE
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Fig. 4. (Left) Syllable segmentation generated by Montréal Forced Aligner [?] opened in Praat [?] for manual correction. (Right) Labeling file generated by the python script. Letters to be highlighted in the Unity interface are manually added.

4 METHODOLOGY

4.1 Participants

This study involved four therapists: two speech therapists (T1, T2) and two physical therapists (T3, T4) (see table 1). All four therapists took part in interviews, while the testing sessions with children were conducted exclusively with T1, a speech therapist specializing in pediatric speech therapy. The children (P1–P8), aged between six and twelve, were already receiving therapy with T1 and had various speech disorders, including autism, Down syndrome, verbal apraxia, and dyslexia (see table 2).

Table 1. Demographic Data of Therapist Participants

Therapist (T)	Gender	Age	Specialty	Years of Experience	Focus Area
T1	Female	35	Speech Therapist	11	Pediatric Speech Pathologies
T2	Female	27	Speech Therapist	4	Post-Stroke Speech Rehabilitation
T3	Female	39	Physiotherapist	14	Motor Rehabilitation
T4	Male	60	Physiotherapist	35	Parkinson's Disease Rehabilitation

Table 2. Demographic Data of Child Participants

Participant (P)	Gender	Age	Pathology
P1	Female	6	Autism
P2	Male	7	Down Syndrome
P3	Female	8	Verbal Apraxia
P4	Male	9	Dyslexia
P5	Female	10	Autism
P6	Male	11	Verbal Apraxia
P7	Female	12	Dyslexia
P8	Male	6	Down Syndrome

4.2 Procedures

The study was structured into three phases: (1) interviews and a demonstration of the rhythmic interface, (2) implementing adaptations based on interview feedback, and (3) testing sessions with the speech therapist and her eight child patients.

4.2.1 Interviews and Demonstration: the interviews with the therapists were divided into three main sections. First, we inquired about their therapeutic practices, specifically how they incorporate speech, rhythm, and gestures into their sessions. Next, we demonstrated the interface to familiarize them with its functionality. Finally, we collected their input on how the interface could be adjusted to better suit their needs. They also provided suggestions on incorporating new content and establishing criteria for evaluating the tool's effectiveness.

Interview questions were predefined as open questions to ensure the collection of a wide range of answers [44]. The responses were recorded and transcribed verbatim for later analysis.

The interview questions were predefined as open questions to ensure the collection of a wide range of responses [44], and they were asked in a consistent manner across all participants to maintain comparability. The questionnaire

used in the interviews can be found in the appendix (9). All responses were recorded and transcribed verbatim for later analysis.

The purpose was to explore the connection between gestures and speech in both speech and physical therapy and to determine how the tool could support both disciplines. This phase also helped us define how we would assess the interface's impact in practice.

4.2.2 Implementation of Adaptations: based on the feedback from the interviews, we made specific changes to the interface to ensure it aligned with the therapists' needs. These adaptations focused both on improving the tool's usability in real-world therapy sessions and on adding exercises tailored to their specific therapeutic practices.

4.2.3 Testing Sessions: the testing sessions were conducted over two days, with T1 supervising eight children (P1-P8) during 30-minute sessions. The children interacted with the adapted interface, and we observed their engagement and progress in speech exercises. T1 provided additional feedback on how the tool functioned in practice and noted any challenges or improvements seen in the children's performance.

4.3 Data Collection and Analysis

Data were collected through several methods to evaluate the interface. All therapist interviews were audio-recorded, transcribed, and analyzed using thematic analysis [8, 9]. This process helped identify patterns related to the integration of the tool into their practices, as well as commonalities between the therapists' approaches, particularly regarding the connection between gestures and speech. This analysis also highlighted areas where the interface could be adapted and improved.

During the testing sessions, observations focused on the children's interactions with the interface, including their engagement, motivation, and progress in speech exercises. The supervising speech therapist (T1) provided feedback on each child's response to the tool. These observations and evaluations were analyzed to understand the interface's impact on the children's performance.

All data were anonymized, and consent was obtained from both therapists and the children's parents. The coding and analysis were conducted by a single researcher and ethical guidelines were followed throughout the study.

5 FINDINGS

5.1 Specific Challenges in Sessions

Rehabilitation sessions in both speech and physical therapy are complex and require constant adaptation. Therapists must adjust their approaches to meet the varying needs and behaviors of patients, which can fluctuate even within a single session. In speech therapy, maintaining a child's engagement is a common challenge, with behavioral responses ranging from hyperactivity to disengagement. T1 noted:

> "It's common to see a child suddenly lose interest and drift away mentally, forcing us to change the session's direction entirely."

T2 emphasized the importance of seizing spontaneous moments:

> "Sometimes, a patient unexpectedly says a word, and you have to seize that moment to build an activity around it."

In physical therapy, the challenges revolve around adjusting exercises to match the patient's fluctuating physical and cognitive states. T3 explained:

> "Even with the same patient, what worked one day might not work the next because of fatigue or changing motivation levels."

Maintaining patient motivation is critical, as T3 pointed out:

> "Patients can quickly become discouraged if they feel they're not progressing, so it's essential to keep them motivated."

These challenges highlight the need for adaptive tools that allow therapists to respond in real-time to patient needs.

5.2 Adaptation Strategies

To enhance therapy effectiveness, therapists use various adaptation strategies to maintain patient engagement and motivation. T2 stressed the value of engaging tools:

> "Educational games and visual aids help keep the session lively and prevent the patient from losing focus."

T1 often uses toys and real-time session adjustments to tailor activities based on the child's reactions. T3 added:

> "If a patient is struggling, you sometimes need to simplify the activity. If they're doing well, you might increase the complexity to keep them challenged."

Rhythm plays a crucial role in therapy, particularly for neurological conditions. T4 discussed using tools like the metronome:

> "We've been using the metronome for a long time, but a constant beat might not always suit real-life scenarios."

For patients struggling with rhythm, devices emitting rhythmic vibrations are useful, as T4 explained:

> "Devices for Parkinson's patients emit vibrations to help maintain their walking rhythm."

Positive reinforcement is another key strategy, as T2 noted:

> "A simple word of encouragement, like 'well done,' can make a big difference in keeping a patient motivated."

Therapists also adjust activities to match patient progress. T3 mentioned using external rhythms in Parkinson's therapy:

> "Patients lose their temporal references and can no longer maintain a rhythm, so we provide an external rhythm that their brain can no longer sustain."

Biofeedback tools are also effective, as T4 stated:

> "Seeing their progress in real-time on a screen motivates patients'."

Tailored approaches are essential, especially in conditions like Parkinson's, where specific tools help maintain functionality. T4 explained:

> "Devices can help Parkinson's patients maintain legible handwriting as it tends to become progressively smaller."

These strategies underscore the importance of flexible, adaptive practices in rehabilitation.

5.3 Interaction Between Motor Gestures and Speech

Motor gestures play a significant role in enhancing speech and communication in therapy. T3 observed:

> "We all talk with our hands. It's spontaneous and helps with communication, even when words are hard to come by."

In speech therapy, gestures reinforce verbal messages and aid memorization. T1 provided an example:

> "I sometimes use objects like small pieces on a table to represent syllables. The child taps each piece as they say the syllable, which helps them see and feel the rhythm."

T1 also noted the effectiveness of combining rhythm and gestures:

> "Tapping out the rhythm while reciting something like the days of the week helps children remember better because they associate the rhythm with the words."

In physical therapy, T3 mentioned using gestures alongside verbal instructions to help patients synchronize actions with therapy rhythms. The integration of speech, gesture, and rhythm is particularly effective for patients with both motor and verbal coordination challenges, enhancing therapeutic outcomes.

6 INTERFACE ADAPTATIONS BASED ON THERAPIST FEEDBACK

Therapist feedback was vital in guiding adaptations to the rhythmic interface, ensuring it meets the diverse needs of patients in both speech and physical therapy. Flexibility and personalization were key priorities.

In response to the need for greater flexibility in therapy sessions, a dedicated web interface was developed, requiring therapists to upload both an audio file (e.g., a word, phrase, or specific exercise, recorded by the therapist or from existing material) and the corresponding text file via a drag-and-drop process. Once the files are uploaded, the interface automatically processes them using the same tools as described in Figure 3, such as Vocal Remover and the Montréal Forced Aligner, to perform segmentation and synchronization of the audio with minimal user input.

In the final stage, the therapist can adjust the automatically generated timestamps by interacting with a visual waveform representation of the audio. The black bars represent calculated timestamps, which can be moved as needed to refine the exercise. Alternatively, the therapist may choose to leave the timestamps unchanged if they are satisfied with the initial output.

The interface includes four main controls: a play button to listen to the current segment (starting with the first segment between timestamps 1 and 2), a next button to skip to the following segment, a third button to play the audio from the current timestamp to the end, and a download button that generates a .txt file. This text file, organized with all the timestamps in the correct format, is ready to be uploaded for patient use in subsequent sessions.

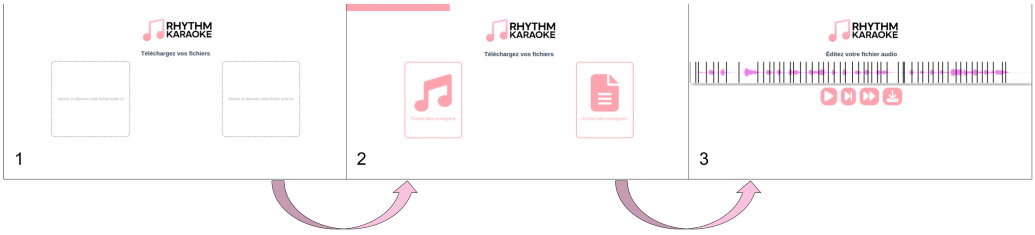

Fig. 5. Workflow of the web interface for uploading and processing audio files. In (1), the drag-and-drop functionality is used to upload audio and text files. In (2), the files are automatically processed. In (3), the interface allows therapists to view and adjust the timestamps (represented as black bars) on the waveform. The control buttons below the waveform enable therapists to navigate through the segments or download the final timestamp file.

The interface now includes a variety of exercises, such as repeating common words or numerical sequences, and integrates words related to daily activities and school terms. Exercises are categorized by speed and length, allowing therapists to adjust difficulty levels, making the tool adaptable to varied patient needs.

User experience improvements were also made, enhancing the interface's intuitiveness and visual clarity. These changes ensure the interface is both engaging for young patients and easy for therapists to integrate into their practices. Overall, the refined rhythmic interface offers a flexible, personalized solution that aligns with current therapeutic needs.

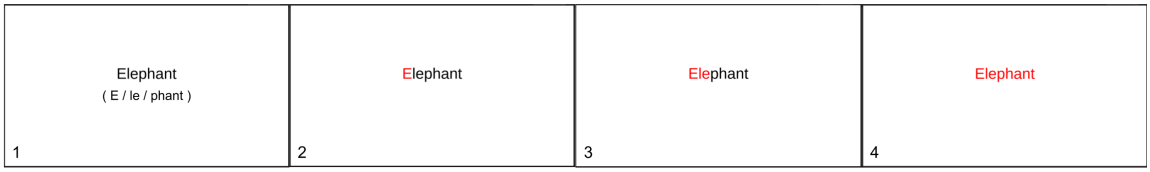

Fig. 6. Illustration of the syllable decomposition and user interactions with the interface during the "Elephant" word repetition exercise. From left to right, images numbered 1 to 4 depict the progressive stages of the exercise. In (1), the word "Elephant" is displayed in black, with no syllables yet pronounced. In (2), after the user presses the button for the first syllable "E," it is highlighted in red, while the remaining syllables stay black. Simultaneously, the synthesized voice pronounces the syllable "E," providing auditory feedback. In (3), the progression continues as the second syllable "le" is pressed, turns red, and is pronounced by the synthesized voice. Finally, in (4), the user completes the exercise by pressing the button for the third syllable "phant," turning the entire word red and triggering the pronunciation of the final syllable. This visual and auditory sequence guides the user through each step, reinforcing coordination between gestures and the pronunciation of syllables, as well as ensuring real-time feedback with each interaction.

7 EXPERIENCE AND EVALUATION WITH CHILDREN

Following interface modifications, therapy sessions were conducted with eight children (Table 2) and T1.

7.1 Motivation and Engagement

During speech therapy sessions, the updated interface quickly captured the attention of P1 to P8, sparking their curiosity. Even the most easily distracted children focused on the screen, exploring its features with enthusiasm and hesitation. The exercises included simple words like "spoon," "apple," and "elephant," 6 as well as word series, poetry excerpts, and phrases from the film *The Jungle Book*.

As words appeared on the screen, the children were fascinated by the accompanying colors and sounds. Some hesitated, gently tapping the screen, while others eagerly pressed the words. T1 observed: "They're curious, but not always patient. Some of them get bored quickly if the word doesn't entertain them enough." This variety in reactions allowed the exercises to be adjusted according to each child's interests.

To better meet the children's needs, additional words were integrated into the interface based on their preferences. For instance, after noticing that P2 responded especially well to animal names, words like "lion," "elephant," and "mouse" were added. T1 explained: "Being able to add words based on their reactions was useful. It allowed us to personalize the sessions even more."

However, while the children responded positively to the system, setting it up required preparation time. Although T1 valued the flexibility of adding new words, this task could be challenging during short sessions. "It's a great tool, but it takes time to set up, and in a 30-minute session, every minute counts," T1 noted.

7.2 Evaluation of Progress

T1 observed mixed results regarding the children's progress. While the interface generally improved concentration and participation, some children struggled to maintain interest over time. "Some kids tune out pretty quickly if the exercise doesn't grab them right away," T1 remarked.

Adding new words helped to re-engage children who were losing focus. For example, when P1 seemed impatient, offering a newly added word could reignite their interest. T1 added: "Adding words based on their interests helped us capture their attention better, even though sometimes it meant juggling with time."

When P3 successfully pronounced a new word, it was particularly rewarding. For instance, P3, who struggled with "apple," eventually managed to say it correctly after several repetitions, boosting their confidence. "Seeing them succeed gives them a boost," T1 commented.

7.3 The Concept of Dose-Effect

T1 emphasized the importance of pacing exercises to avoid exhausting the children's attention. "Too many exercises in a row, and they check out," T1 observed. The interface allowed for structuring sessions into short, repeated sequences while staying attentive to signs of fatigue or disinterest.

Short but regular exercises, combined with the gradual introduction of new words, were favored to help the children practice without feeling overwhelmed. T1 noted: "Introducing a new word mid-session often re-engages their attention."

However, managing this flexibility in short sessions was challenging. T1 needed to balance the time spent configuring the interface with direct interaction. "It's great to personalize, but you have to find a balance so it doesn't take up too much session time," T1 mentioned.

7.4 Impact on Verbal Skills

The interface positively impacted the children's verbal and motor skills, though results varied. Some children repeated words with precision, while others rushed, sometimes without correct pronunciation. "Some repeat just to finish faster," T1 observed, stressing the importance of encouraging quality over quantity in repetitions.

In some cases, adding new words provided additional challenges, helping children progress further. For example, P8, who struggled with certain sounds, showed significant improvement after more complex words were introduced, allowing for targeted work on specific phonemes.

In conclusion, the experience with P1 to P8 demonstrated that the rhythmic interface is a promising tool for improving motivation and language skills. While adjustments are needed to match each child's pace and reactions, and while adding new words requires time, the observed progress in participation and linguistic performance highlights the potential of this technology to enrich therapeutic practices. A flexible approach tailored to individual needs is key to maximizing the tool's effectiveness while managing session time efficiently.

8 DISCUSSION

Our study confirmed some anticipated outcomes while revealing unexpected insights, particularly regarding the cross-domain potential of integrating rhythm, gestures, and speech in therapy. As expected, the rhythmic interface enhanced children's motivation and engagement, consistent with existing research emphasizing rhythm's role in sustaining attention and facilitating learning. The ability to customize exercises by adding new words based on individual preferences significantly improved the flexibility and efficacy of the sessions, allowing therapists to adapt more precisely to each child's needs.

One of the key findings was the successful adaptation of a motor rehabilitation tool to speech therapy, taking advantage of the cognitive link between gestures and speech. This discovery aligns with research by McNeill [33], which demonstrated how gestures are integral to language processing. Our findings extend this understanding, showing that a system initially designed to enhance motor skills can also be applied to speech rehabilitation, leveraging the link between gesture and verbal communication.

An additional insight from our results was the influence of rhythm on speech fluency. The controlled tempo provided by the interface facilitated smoother speech production, consistent with findings on slow speech rate techniques used in stuttering therapy [21, 28]. This technique, by reducing cognitive load, likely allowed children more time to articulate their speech more clearly, leading to improved verbal output.

The multisensory nature of the interface, which combines auditory, visual, and motor stimuli, further amplified

the children's engagement. Studies have demonstrated that such multisensory inputs enhance both learning and memory retention by stimulating multiple brain regions simultaneously [30]. This capability to engage different senses in a synchronized manner likely played a key role in maintaining the children's focus and facilitating their learning process.

The interactive nature of the system, enhanced by real-time feedback, provided a powerful reinforcement mechanism. Research shows that immediate auditory and visual feedback can accelerate learning by reinforcing the association between actions and outcomes [46, 47]. The interface's feedback mechanism likely helped the children solidify the connection between gestures and verbal expressions, improving their overall speech performance.

One notable result was the increased verbal output observed in children using the interface. The combination of interactivity and personalization, allowing for tailored exercises, may have encouraged greater participation. This aligns with Asher's work [3], which shows that incorporating physical movement into language learning stimulates verbal production. The rhythmic and interactive nature of our system may have created an environment conducive to more spontaneous speech.

In terms of long-term benefits, research suggests that rhythmic and motor-based interventions can have lasting effects on neural connectivity, supporting sustained improvements in both motor and verbal skills [46, 47]. Investigating whether the progress observed during therapy sessions translates into durable gains in language acquisition and fluency would be a valuable direction for future research.

Given these findings, the potential to apply this system in reverse—using speech to enhance motor skills—deserves exploration. For instance, verbal cues could be incorporated into motor rehabilitation exercises to strengthen the cognitive link between speech and movement, potentially improving outcomes in stroke recovery or neurodegenerative diseases.

Moreover, the use of rhythmic gestures could be expanded into language learning environments. The coordination of speech and movement might facilitate vocabulary acquisition and reinforce cognitive associations, as demonstrated in previous studies on the role of motor activities in language learning [30]. Our findings suggest that gesture-based rhythmic interfaces could be developed to assist learners by synchronizing vocabulary learning with physical actions, amplifying the benefits observed in methods like the Total Physical Response approach [3].

In terms of rehabilitation, our study supports the idea that integrating rhythm and gestures is beneficial across various conditions, including aphasia, where patients face both motor and verbal coordination challenges. Prior research has shown that gestures can aid speech recovery in aphasia patients [48]. The rhythmic interface could be further adapted to make therapy more interactive and engaging, where patients use gestures to trigger words, strengthening the connection between movement and language. Additionally, this approach could be explored in contexts of neurostimulation, where the interaction between rhythmic stimulation and speech production remains under-researched [46, 47].

Finally, the connection between gestures and speech could open new avenues in neuroeducation, particularly in managing conditions like attention-deficit hyperactivity disorder. Integrating rhythmic gestures into cognitive exercises

could help channel children's motor energy while enhancing focus and learning, as supported by studies on the impact of physical activity on executive functions [15].

9 CONCLUSIONS

Our research explores the adaptation of a rhythmic interface initially designed for manual dexterity rehabilitation, which we successfully applied to speech therapy. This adaptation reveals the cognitive link between gestures and speech, enhancing motivation and therapeutic engagement in children with speech disorders. Our study contributes to understanding the challenges of speech therapy with children and can guide designers interested in incorporating novel technologies in this field.

The study opens up possibilities for cross-domain applications, including speech improvement, motor skills rehabilitation in neurorehabilitation contexts like stroke recovery, and educational settings where synchronized gestures and speech could aid language learning and cognitive development.

Preliminary results suggest that the rhythmic interface can strengthen the connection between gestures and verbal communication in speech therapy. Further studies are needed to validate these findings in broader, long-term therapeutic contexts.

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APPENDIX A: INTERVIEW PROTOCOL

Introduction to the Interview

I am currently developing a rhythm-based system that allows users to control the rhythm of synthesized vocal phrases through finger tapping. The system is being designed for potential use in therapeutic settings, with the aim of supporting both motor rehabilitation and speech therapy by incorporating rhythmic exercises. The purpose of this interview is to gather your professional insights on how such a system could be integrated into existing therapeutic practices and to understand your perspective on the potential benefits or challenges it may present. Additionally, I am interested in learning more about the current therapeutic methods you use, to better inform the system's development and ensure it aligns with the practical needs of clinicians and patients.

Part 1: Questions about Current Practice

(1) How are speech, rhythm, and gestures currently integrated into your practice?
(2) What types of content do you typically use during your sessions?
(3) What challenges do you encounter in your sessions related to speech, rhythm, and gestures?

Part 2: Demonstration

The demonstration of the interface takes place in this part of the interview.

Part 3: Questions about the Interface

A: Interface Feedback and Current Use.

(1) What are your thoughts on the interface? What do you see as its advantages and disadvantages?
(2) Can you provide examples of specific situations where you think this system could be particularly beneficial for patients?
(3) Can you think of other types of content that could be integrated into this system to improve it?

B: Evaluation and Testing with Patients.

(1) Would you be interested in testing this system with your clients/patients to evaluate its effectiveness and impact in a real-world context? If so, what steps and criteria would you suggest for conducting such a test?
(2) What success indicators or metrics would you consider most relevant for evaluating the effectiveness of this system?
(3) Were the questions I asked relevant? Do you feel that anything was missing? Do you have any additional comments or suggestions?

A APPENDIX B: IDENTIFIED CHALLENGES AND ADAPTIVE STRATEGIES IN REHABILITATION SESSIONS

Theme	Specific Challenge	What Worked Well	What Did Not Work	Adaptation Strategy
Speech Therapy	Maintaining child engagement despite distractions	Seizing spontaneous moments of speech (e.g., when the child unexpectedly speaks)	Sudden loss of focus, hyperactivity, disengagement mid-session	Real-time adjustments using toys and educational games to maintain engagement. Use of rhythm and visual aids for consistency.
	Handling wide range of behavioral responses (hyperactivity to disengagement)	Adjusting activities based on child's reaction, making session dynamic and flexible	Some methods losing effectiveness mid-session if child loses focus or gets bored	Dynamic shift in session, introducing new stimuli like toys or objects representing syllables to re-engage the child.
	Using rhythm and gestures to reinforce speech	Combining rhythm with gestures (e.g., tapping objects to syllables) to improve memory	Repetitive rhythm may become less effective if overused in long sessions	Incorporating both auditory (rhythm) and physical (gestures) cues to reinforce learning, allowing breaks to re-introduce focus.
Physical Therapy	Adjusting exercises to fluctuating physical and cognitive states (fatigue, motivation)	Simplifying or complicating exercises in real time based on patient performance	Exercises too difficult when patients are tired, leading to discouragement	Constant adaptation by simplifying when necessary, using positive feedback to maintain motivation.
	Keeping patients motivated throughout sessions	Positive reinforcement (e.g., verbal encouragement) and use of external rhythmic cues (e.g., metronome, devices for Parkinson's)	Patients becoming demotivated if they don't perceive progress	Providing biofeedback tools showing real-time progress, helping patients visually understand their improvements.
	Balancing cognitive load with physical tasks	Breaking down complex exercises into smaller, manageable parts	Overloading patients with tasks that are too complex or lengthy	Simplifying instructions, providing extra time for task transitions, reinforcing small victories.
Motor-Speech Integration	Using motor gestures to aid verbal expression	Gestures (e.g., hand movements) help reinforce and enhance verbal memory (linking movement to rhythm)	Difficulty synchronizing speech and motor gestures in some patients with advanced conditions	Providing physical objects to represent concepts (e.g., syllables) and using rhythm to support coordination.
	Synchronizing verbal instructions with physical tasks	Combining verbal cues with synchronized motor actions improves speech recall	Unsynchronized gestures can lead to confusion or failure to recall verbal cues	Breaking down tasks, offering more time for synchronization, and giving real-time feedback during sessions.
General Therapy Strategies	Adjusting methods to different patient needs during the session	Real-time adaptability of tools, switching between methods depending on patient response	Rigid tools without adaptive features can impede flexibility	Using adaptive tools like metronomes with variable rhythms, biofeedback for real-time progress tracking.
	Incorporating rhythm in therapy for neurological conditions (e.g., Parkinson's)	Devices emitting rhythmic vibrations to help patients maintain walking or other activities	Monotonous rhythms may fail to engage patients long-term	Using devices that emit varied rhythmic cues, offering flexibility depending on patient's needs (e.g., adjusting speed or intensity).
	Adapting educational tools in therapy	Interactive, engaging tools (e.g., games, toys) help maintain focus and motivation	Standard tools become ineffective if they don't adapt to patient progression or mood changes	Use of customizable, interactive tools that adapt to the patient's real-time reactions and needs during therapy.

Table 3. Challenges, successful strategies, and adaptations observed during speech and physical therapy sessions, focusing on real-time adjustments and adaptive techniques used by therapists.

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